Abdullah Abu Zaid, PhD

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Research Interests

AI-enabled non-terrestrial networks; HAPS and UAV communications; foundation models and LLM agents for network automation; stochastic geometry and performance analysis of aerial-terrestrial networks; learning-based network optimization; aerial perception for autonomous missions.

Education

King Abdullah University of Science and Technology — Thuwal, Saudi Arabia
PhD in Electrical and Computer Engineering, Advisor: Prof. Mohamed-Slim Alouini — *04/2022 – 12/2025*

- Dissertation: *"Urban Air Mobility Communications: Performance Analysis and Design"*

Georgia Institute of Technology — Remote
*MS in Computer Science, Specialization in Machine Learning, GPA: **3.90/4.0*** — *08/2020 – 12/2024*

- Graduate coursework: Deep Learning, Computer Vision, Machine Learning, Artificial Intelligence, HPC

King Abdullah University of Science and Technology — Thuwal, Saudi Arabia
*MS in Electrical and Computer Engineering, GPA: **3.73/4.0*** — *08/2020 – 04/2022*

University of Jordan — Amman, Jordan
*BSc in Electrical Engineering, **Ranked 1st/81**, GPA: **3.80/4.0*** — *10/2015 – 06/2020*

Experience

Researcher — Thuwal, Saudi Arabia
King Abdullah University of Science and Technology, Communication Theory Lab — *08/2020 – Present*

- Designed and simulated an AI-enabled HAPS–UAV search-and-rescue system for remote desert operations, coupling a fine-tuned YOLO detector with a network simulator to evaluate autonomous mission and transmission policies against ground-team baselines. Awarded 3rd place in IEEE/CST NTN competition (2026).
- Developed stochastic-geometry and Monte Carlo models for UAV, eVTOL, and HAPS networks, analyzing coverage, line-of-sight probability, and link reliability for aerial platforms operating over urban and remote terrain (4 IEEE journal publications: TAES, TVT, OJVT, Communications Magazine).
- Built a road-aware line-of-sight probability model for low-altitude aerial vehicles, awarded Best Paper by ICAO in a global competition of 100 entries.
- Implemented large-scale Python and MATLAB simulation frameworks for aerial-terrestrial heterogeneous networks, including mmWave and NOMA-based aerial-assisted vehicular links, and ran them as parallel batch jobs on KAUST's Ibex HPC cluster (Slurm) to quantify throughput and outage under realistic 3D deployments.
- Presented research at IEEE ICC, EuCNC/6G Summit, and CTW; placed 3rd in the IEEE 6G & NTN Contest (2023) and IEEE ICC Four-Minute Thesis (2025).

Autonomous Systems Intern — Riyadh, Saudi Arabia
Boeing — *10/2025 – 03/2026*

- Contributed to efforts evaluating the integration of AI and ML into Boeing's autonomous systems and aerospace programs, working with distributed teams across Boeing's U.S. and international sites to identify candidate applications, expected benefits, and implementation challenges.
- Reviewed and provided technical feedback on the Boeing–SkyGrid–Wisk *Concept of Operations for Automated Flight Rules (AFR)*, a publicly released framework for integrating highly automated and uncrewed aircraft into shared airspace alongside VFR/IFR, covering detect-and-avoid, automated separation, and mission-management automation.

Teaching Assistant — Thuwal, Saudi Arabia
King Abdullah University of Science and Technology — *01/2022 – 05/2022*

- Assisted in teaching, grading assignments, and held office hours for approximately 40 students.

IAESTE Research Internship — Lleida, Spain
University of Lleida, Polytechnic School, Mathematics Department — *06/2019 – 07/2019*

- Implemented popular cryptosystem algorithms such as RSA and DL using Python and SageMath.

Publications

- Extended Road-Aware Line-of-Sight Probability Model for Urban Air Mobility
  IEEE Transactions on Aerospace and Electronic Systems, 2025. Paper Link
  Best paper award at *Global AAM, ICAO, 2024, Montreal, Canada.*

- Aerial-Terrestrial Heterogeneous Networks for Urban Air Mobility: A Performance Analysis
  IEEE Open Journal of Vehicular Technology, 2025. Paper Link
  Poster at *2024 IEEE CTW, Banff, Canada.*

- eVTOL Communications and Networking in UAM: Requirements, Key Enablers, and Challenges
  IEEE Communications Magazine, 2023. Paper Link
  Poster at *2023 IEEE EuCNC and 6G Summit, Gothenburg, Sweden.*

- Aerial-aided mmWave VANETs using NOMA: Performance Analysis, Comparison, and Insights
  IEEE Transactions on Vehicular Technology, 2023. Paper Link
  Poster at *2022 IEEE EuCNC and 6G Summit, Grenoble, France.*

Awards, Honors, and Fellowships

- **3rd Place** out of 81 entries, $5,000 award, CST/IEEE Non-Terrestrial Networks (NTN) 2026 International Competition, Riyadh, Saudi Arabia, October 2026. Competition Link
- **KAUST Graduate Fellowship** — full tuition, stipend, and research support, 2020–2025
- **Dean's List Award**: $2,500 award for top students, CEMSE Division, KAUST, 2025
- **3rd Place**, Four Minute Thesis (4MT) Competition, IEEE International Conference on Communications (ICC), Montreal, Canada, 2025. Presentation Link
- **Best Paper Award** (1 of 100 entries), Global Advanced Air Mobility (AAM) Academic Paper Competition, International Civil Aviation Organization (ICAO), Montreal, Canada, 2024. ICAO Link
- **3rd Place**, IEEE 6G and NTN: Challenges and Solutions Contest, 2023. IEEE Link
- **Ranked 1st/81**, BSc in Electrical Engineering, University of Jordan, 2020

Professional Service

Journal Peer Reviewer: IEEE Transactions on Vehicular Technology, IEEE Transactions on Industrial Informatics, IEEE Communications Magazine.
Conference Peer Reviewer: IEEE VTC2023-Spring (97th), IEEE VTC2024-Spring (99th), IEEE PIMRC 2025.
University Service, KAUST: Orientation Leader (2024, 2025); Winter Enrichment Program (2023, 2025); Career Fair (2022, 2023); Commencement (2023).

Technical Skills

Programming: Python, MATLAB, C++, SQL, CUDA
Machine Learning: PyTorch, TensorFlow, scikit-learn, CV, LLMs, RAG
Simulation & Computing: Monte Carlo and stochastic-geometry simulation, network simulation, SLURM/HPC clusters (KAUST Ibex), parallel and distributed computing, Linux, Docker, Git
Cloud: AWS (EC2, S3, Lambda); AWS Certified Solutions Architect – Associate

Languages

Arabic: Native
English: Professional proficiency (IELTS 8.0; TOEFL iBT 105)
French: Basic proficiency (A1)
German: Basic proficiency (A1)